

Department of Leather Engineering
Khulna University of Engineering & Technology
B. Sc. Engineering 2nd Year 1st Examination-2020
Basic Mechanical Engineering
ME 2119

Time: 1.5 Hours

Full Marks: 120

N.B. i) Answer any two questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION-A

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|---|----|
| 1(a) Define a thermodynamic system, boundary and surrounding. | 06 |
| 1(b) State and explain Zeroth law of thermodynamics. Why this law is considered as the basis for temperature measurement? | 10 |
| 1(c) Law of conservation of energy can be derived from a law of thermodynamics, show the derivation. | 14 |
| 2(a) Heat cannot be transferred from a cold body to a hot body without consuming external energy, briefly explain it with the aid of thermodynamic law. | 12 |
| 2(b) What are the limitations of 1 st law of thermodynamics? Explain. | 06 |
| 2(c) 6.5 kg of a gas expands within a flexible container so that the pressure-volume relation is of the form $PV^{1.39} = \text{Constant}$. The initial pressure is 1.0 MPa and the final pressure is 5 KPa. The initial volume is 1.5 m ³ . The specific internal energy of the gas decreases by 41 KJ. Find the heat transfer in magnitude and direction. | 12 |
| 3(a) Draw the P-V and T-S diagrams of both the Otto and Diesel cycle, and compare the thermodynamic processes. Write down their efficiency equations and compare them for same compression ratio. | 15 |
| 3(b) Briefly explain the working principle of Carnot cycle with necessary diagrams. Mention, why it is not possible to make an engine working on Carnot's cycle. | 15 |

SECTION-B

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|---|----|
| 4(a) With neat sketch explain how reciprocating motion is converted to rotary motion in an IC engine. | 09 |
| 4(b) Explain with necessary diagrams the sequences of operations of a four stroke cycle diesel engine. | 12 |
| 4(c) Differentiate between two stroke cycle engine and four stroke cycle engine. | 09 |
| 5(a) Compare the prime features of a fire tube boiler and a water tube boiler. | 12 |
| 5(b) Write briefly notes on (i) Water level indicator; (ii) Fusible plug; (iii) Air preheater; (iv) Super heater. | 12 |

- 5(c) Differentiate between boiler mountings and boiler accessories. 06
- 6(a) A steam plant consisting of a boiler, economizer and a superheater, has the following particulars. 18
 Steam pressure = 14 bars; Mass of steam generate = 4500 kg/h, Mass of coal used = 635 kg/h; Calorific value of coal = 29,400 KJ/kg of coal; Temperature of feed water entering the economizer = 30°C; Temperature of feed water leaving the economizer = 130°C; Dryness fraction of steam leaving the boiler = 0.96; Temperature of steam leaving the superheater = 320°C.
 Determine: (i) Overall efficiency of the plant, and (ii) The percentage of the available heat utilized in the boiler, economizer and superheater.
 # Given data from the steam table:
 $h_{f1@30^{\circ}C} = 125.7 \text{ KJ/kg}$; $h_{f2@130^{\circ}C} = 546.3 \text{ KJ/kg}$; @14bar; $h_f = 830.1 \text{ KJ/kg}$; $h_{fg} = 1957.7 \text{ KJ/kg}$; $h_g = 2787.8 \text{ KJ/kg}$; $t = 195^{\circ}C$
- 6(b) Briefly explain the working principle of a impulse turbine. 12

Department of Leather Engineering
Khulna University of Engineering & Technology
B. Sc. Engineering 2nd Year 1st Term Examination-2020
Accounting and Industrial Law
Hum 2119

Time: 1.5 Hours

Full Marks: 120

- N.B i) Answer any **TWO** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION-A

- 1(a) Define accounting. State the importance of accounting in modern business. 15
1(b) What is accounting equation? Explain its elements. 15

- 2(a) What are the characteristics of transactions? 05
2(b) On February 01, 2021 Mr. Kabir established Pioneer Travel Agency. The following 25
transactions were completed during the month:

- February 01: Owner brought cash to start the agency Tk. 2,95,000.
03: Purchased office equipment for cash Tk. 34,000.
08: Service revenue performed of Tk. 37,000, of which received cash Tk. 15,000 and balance on billed.
10: Advertisement expense paid in cash Tk. 4,800.
13: Paper and other supplies purchased on account Tk. 10,500 from M/S Surma Traders.
15: Life insurance premium paid in cash Tk. 5,200 from travel agency.
18: Cash collection from customer Tk. 12,500 related on February 08.
24: Hired an expert manager at a salary of Tk. 18,000 per month, effective March 01, 2021.
25: Paid cash Tk. 4,500 to M/S Surma Traders.
28: Utilities expenses paid in cash Tk. 9,600.

Required: Show journal entries (without explanation) in the books of Pioneer Travel Agency.

- 3(a) Kazi Salim Hayder started his law practice on March 01, 2021. His other transactions 30
during the month were as follows:

- March 01: He invested cash Tk. 1,65,000 in the law firm.
" 02: Paid cash Tk. 4,000 for rent of the month.
" 05: Office supplies purchased on credit Tk. 11,000.
" 09: Service rendered for cash Tk. 23,000.
" 15: Borrowed Tk. 48,000 cash from a bank on a note payable.
" 18: Cash paid Tk. 5,000 due on march 05.
" 20: Service revenue performed on account Tk. 27,500.
" 28: Received Tk. 16,000 in cash from customer who have previously serviced on March 20.
" 31: Office supplies used Tk. 7,500 during the month.

Instruction: Show the effect of transactions on accounting equation in tabular form.

SECTION-B

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| 4(a) | Define industrial law. Who sets this law and for what purposes? | 15 |
| 4(b) | As per Bangladesh Labor Act, 2006 – Who is liable to pay compensation to workers for an injury caused by accident? Under what conditions that person will not be held responsible? | 15 |
| 5(a) | What is meant by wages? Write down the provisions regarding deductions that may be made from wages. | 20 |
| 5(b) | How unpaid wages of a dead worker is to be paid? | 10 |
| 6(a) | What is ILO? Write down the principles of ILO. | 10 |
| 6(b) | State the objectives of ILO. | 10 |
| 6(c) | Describe the functions of labour court. | 10 |

Department of Leather Engineering
Khulna University of Engineering & Technology
 B. Sc. Engineering 2nd Year 1st Examination-2020
Leather Manufacturing Technology II
LE 2101

Time: 1.5 Hours

Full Marks: 120

- N.B. i) Answer any two questions from each section in separate scripts.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable data if any missing.

SECTION-A

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| 1(a) | Describe dewooling and liming. Give the enzymatic method of unhairing with advantages. | 08 |
| 1(b) | Illustrate with neat sketch the hair/wool destruction in very strong liming conditions. | 06 |
| 1(c) | State and define the types of unhairing. Assume you are given a freshly flayed goat skin; make a recipe for your goat skin to conduct preservation (wet-salting) pre-soaking, soaking and painting. | 11 |
| 1(d) | Briefly discuss the chemistry of liming. | 05 |
| 2(a) | Describe lime and reliming. Concisely discuss the three pit liming method with figures. | 08 |
| 2(b) | State the significance of dermatan sulphate removal during liming operation. | 08 |
| 2(c) | Discuss the behavior of collagen in acidic and alkaline media. | 06 |
| 2(d) | Write short notes: (i) Swelling and plumping and (ii) Lime blast | 08 |
| 3(a) | Deliming is necessary before chrome tanning – Explain. | 05 |
| 3(b) | Write down the benefit of limed fleshing. Explain the environmental impact of fleshing. | 08 |
| 3(c) | Write down the mechanism of ammonium salt based deliming with advantages and environmental effects. | 08 |
| 3(d) | Suppose you are given a wet salted goatskin (wt. 1.75 kg) after liming pelt weight was increased 0.45% (by wt.). Calculate the amount of ammonium chloride to be needed for deliming to remove 50% lime. Assume, after lime wash, pelt was content 1.15% calcium hydroxide. | 09 |

SECTION-B

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|------|--|----|
| 4(a) | State the role of enzymes in bating operation of leather processing. | 08 |
| 4(b) | Give the effect of the presence of excess natural grease in animal skins from leather manufacturing point of view. | 08 |
| 4(c) | Write the mechanism of aqueous emulsification method in skin degeasing. | 09 |
| 4(d) | Acid bate has advantages over conventional bate – Explain. | 05 |
| 5(a) | “Acidification of pelt before mineral tanning is must”-Justify the statement. | 07 |
| 5(b) | Write notes on the following terms: (i) Pickling additives and (ii) Pickling salts | 08 |
| 5(c) | State the role of non swelling acids in pickling. Give some examples of non swelling acids. | 07 |

5(d)	Make a single question on depickling of hides and skins and give the answer. The question should be in such a way that, all the aspects of depickling operation will come under it.	08
6(a)	Give the mechanism of interaction of olated and oxolated chrome complexes with skin collagen.	06
6(b)	State the principle of single bath chrome tanning process.	08
6(c)	Define masking agent. Write down the role of masking in chrome tanning.	09
6(d)	Calculate the formula of 50% basic chrome sulphate.	07